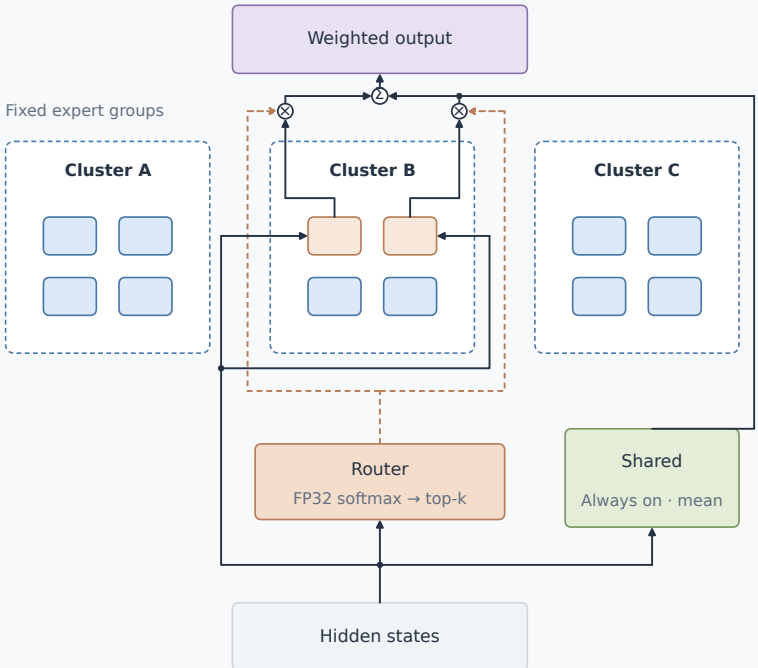
Chronos v3

Routing locality and expert communities

a SPARSE ROUTING WITH SHARED RESIDUAL



b TRAINING OBJECTIVE

Stream / document

Aggregate all dispatch probabilities

Routing-locality objective

Concentrate within a stream
spread mass across streams

$$\mathcal{L}_{\text{cl}} = 1 - \frac{I(w;c)}{\log N_c}$$

Fixed ID partition; learned routing.
Not physical migration of experts.

Enabled only when its coefficient > 0.

Communities here are a conceptual view of routing specialization; positions and colours are not measured embeddings.

Offline alternative: activation traces → coactivation graph → Louvain / greedy partition → cluster-page packing.

Training partition and offline graph partition are distinct mechanisms; neither implies spontaneous expert motion.

The locality objective encourages reuse of expert pages across tokens and recurrent dispatches; no speedup is asserted.